

1. Consider a theory with the interaction given by $L = ig\bar{\psi}\gamma_5\psi\pi$. It represents the interaction of electrons (e^-) and positrons (e^+) with a real pseudoscalar field π . ψ denotes the fermionic field operator. Write down the Feynman rules of the theory and using these to lowest nontrivial order draw all diagrams and write down the corresponding S matrix elements for the following processes: (1) $e^- + \pi \rightarrow e^- + \pi$, (2) $e^+ + \pi \rightarrow e^+ + \pi$, (3) $e^- + e^- \rightarrow e^- + e^-$, (4) $e^- + e^+ \rightarrow e^- + e^+$.
2. A scalar field ϕ and a pseudoscalar field π interact with electrons and positrons through the following interaction: $L = g\bar{\psi}\psi\phi + ig'\bar{\psi}\gamma_5\psi\pi$. To lowest nontrivial order draw all diagrams and write down the corresponding S matrix elements for the following processes: (1) $e^- + \phi \rightarrow e^- + \pi$, (2) $e^+ + e^- \rightarrow \pi + \phi$, (3) $\pi + \phi \rightarrow e^+ + e^-$, (4) $e^+ + \pi \rightarrow e^+ + \phi$.